Data Cleaning with Hotel Booking Dataset

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Background

In this notebook, Weâ€™ll review a scenario, and focus on cleaning real data in R. We will use data cleaning functions and perform basic calculations to gain initial insights into our data.

The scenario

In this scenario, we are junior data analysts working for a hotel booking company. We have been asked to clean a .csv file that was created after querying a database to combine two different tables from different hotels. In order to learn more about this data, We are going to need to use functions to preview the data's structure, including its columns and rows. We will also need to use basic cleaning functions to prepare this data for analysis.

Step 1: Load packages

Start by installing the required packages.

```
#install.packages("tidyverse")
#install.packages("skimr")
#install.packages("janitor")
```

load libraries:

```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.1.4     ✔ readr     2.1.4
## ✔ forcats   1.0.0     ✔ stringr   1.5.1
## ✔ ggplot2   3.4.4     ✔ tibble    3.2.1
## ✔ lubridate 1.9.3     ✔ tidyr     1.3.0
## ✔ purrr     1.0.2
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to bec
ome errors
```

```
library(skimr)
library(janitor)
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

Step 2: Import data

The data in this example is originally from the article Hotel Booking Demand Datasets (https://www.sciencedirect.com/science/article/pii/S2352340918315191 (https://www.sciencedirect.com/science/article/pii/S2352340918315191)), written by Nuno Antonio, Ana Almeida, and Luis Nunes for Data in Brief, Volume 22, February 2019.

The data was downloaded and cleaned by Thomas Mock and Antoine Bichat for #TidyTuesday during the week of February 11th, 2020 (https://github.com/rfordatascience/tidytuesday/blob/master/data/2020/2020-02-11/readme.md (https://github.com/rfordatascience/tidytuesday/blob/master/data/2020/2020-02-11/readme.md)).

We can learn more about the dataset here: https://www.kaggle.com/jessemostipak/hotel-booking-demand (https://www.kaggle.com/jessemostipak/hotel-booking-demand)

Import data from a .csv in the project folder called "hotel_bookings.csv" and save it as a data frame called `bookings_df` :

```
bookings_df <- read_csv("hotel_bookings.csv")
```

```
## Rows: 119390 Columns: 32
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## chr  (13): hotel, arrival_date_month, meal, country, market_segment, distrib...
## dbl  (18): is_canceled, lead_time, arrival_date_year, arrival_date_week_numb...
## date  (1): reservation_status_date
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Step 3: Getting to know our data

Before We start cleaning our data, take some time to explore it. We can use several functions to preview our data:

```
head(bookings_df)
```

```
## # A tibble: 6 × 32
##   hotel        is_canceled lead_time arrival_date_year arrival_date_month
##   <chr>              <dbl>     <dbl>             <dbl> <chr>
## 1 Resort Hotel           0       342              2015 July
## 2 Resort Hotel           0       737              2015 July
## 3 Resort Hotel           0         7              2015 July
## 4 Resort Hotel           0        13              2015 July
## 5 Resort Hotel           0        14              2015 July
## 6 Resort Hotel           0        14              2015 July
## # ℹ 27 more variables: arrival_date_week_number <dbl>,
## #   arrival_date_day_of_month <dbl>, stays_in_weekend_nights <dbl>,
## #   stays_in_week_nights <dbl>, adults <dbl>, children <dbl>, babies <dbl>,
## #   meal <chr>, country <chr>, market_segment <chr>,
## #   distribution_channel <chr>, is_repeated_guest <dbl>,
## #   previous_cancellations <dbl>, previous_bookings_not_canceled <dbl>,
## #   reserved_room_type <chr>, assigned_room_type <chr>, …
```

We can summarize or preview the data with the `str()` and `glimpse()` functions to get a better understanding of the data

```
str(bookings_df)
```

```
## spc_tbl_ [119,390 × 32] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ hotel                         : chr [1:119390] "Resort Hotel" "Resort Hotel" "Resort Hote
l" "Resort Hotel" ...
##  $ is_canceled                   : num [1:119390] 0 0 0 0 0 0 0 0 1 1 ...
##  $ lead_time                     : num [1:119390] 342 737 7 13 14 14 0 9 85 75 ...
##  $ arrival_date_year             : num [1:119390] 2015 2015 2015 2015 2015 ...
##  $ arrival_date_month            : chr [1:119390] "July" "July" "July" "July" ...
##  $ arrival_date_week_number      : num [1:119390] 27 27 27 27 27 27 27 27 27 27 ...
##  $ arrival_date_day_of_month     : num [1:119390] 1 1 1 1 1 1 1 1 1 1 ...
##  $ stays_in_weekend_nights       : num [1:119390] 0 0 0 0 0 0 0 0 0 0 ...
##  $ stays_in_week_nights          : num [1:119390] 0 0 1 1 2 2 2 2 3 3 ...
##  $ adults                        : num [1:119390] 2 2 1 1 2 2 2 2 2 2 ...
##  $ children                      : num [1:119390] 0 0 0 0 0 0 0 0 0 0 ...
##  $ babies                        : num [1:119390] 0 0 0 0 0 0 0 0 0 0 ...
##  $ meal                          : chr [1:119390] "BB" "BB" "BB" "BB" ...
##  $ country                       : chr [1:119390] "PRT" "PRT" "GBR" "GBR" ...
##  $ market_segment                : chr [1:119390] "Direct" "Direct" "Direct" "Corporate" ...
##  $ distribution_channel          : chr [1:119390] "Direct" "Direct" "Direct" "Corporate" ...
##  $ is_repeated_guest             : num [1:119390] 0 0 0 0 0 0 0 0 0 0 ...
##  $ previous_cancellations        : num [1:119390] 0 0 0 0 0 0 0 0 0 0 ...
##  $ previous_bookings_not_canceled: num [1:119390] 0 0 0 0 0 0 0 0 0 0 ...
##  $ reserved_room_type            : chr [1:119390] "C" "C" "A" "A" ...
##  $ assigned_room_type            : chr [1:119390] "C" "C" "C" "A" ...
##  $ booking_changes               : num [1:119390] 3 4 0 0 0 0 0 0 0 0 ...
##  $ deposit_type                  : chr [1:119390] "No Deposit" "No Deposit" "No Deposit" "No
Deposit" ...
##  $ agent                         : chr [1:119390] "NULL" "NULL" "NULL" "304" ...
##  $ company                       : chr [1:119390] "NULL" "NULL" "NULL" "NULL" ...
##  $ days_in_waiting_list          : num [1:119390] 0 0 0 0 0 0 0 0 0 0 ...
##  $ customer_type                 : chr [1:119390] "Transient" "Transient" "Transient" "Trans
ient" ...
##  $ adr                           : num [1:119390] 0 0 75 75 98 ...
##  $ required_car_parking_spaces   : num [1:119390] 0 0 0 0 0 0 0 0 0 0 ...
##  $ total_of_special_requests     : num [1:119390] 0 0 0 0 1 1 0 1 1 0 ...
##  $ reservation_status            : chr [1:119390] "Check-Out" "Check-Out" "Check-Out" "Check
-Out" ...
##  $ reservation_status_date       : Date[1:119390], format: "2015-07-01" "2015-07-01" ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   hotel = col_character(),
##   ..   is_canceled = col_double(),
##   ..   lead_time = col_double(),
##   ..   arrival_date_year = col_double(),
##   ..   arrival_date_month = col_character(),
##   ..   arrival_date_week_number = col_double(),
##   ..   arrival_date_day_of_month = col_double(),
##   ..   stays_in_weekend_nights = col_double(),
##   ..   stays_in_week_nights = col_double(),
##   ..   adults = col_double(),
##   ..   children = col_double(),
##   ..   babies = col_double(),
##   ..   meal = col_character(),
##   ..   country = col_character(),
##   ..   market_segment = col_character(),
##   ..   distribution_channel = col_character(),
##   ..   is_repeated_guest = col_double(),
```

```
##   ..   previous_cancellations = col_double(),
##   ..   previous_bookings_not_canceled = col_double(),
##   ..   reserved_room_type = col_character(),
##   ..   assigned_room_type = col_character(),
##   ..   booking_changes = col_double(),
##   ..   deposit_type = col_character(),
##   ..   agent = col_character(),
##   ..   company = col_character(),
##   ..   days_in_waiting_list = col_double(),
##   ..   customer_type = col_character(),
##   ..   adr = col_double(),
##   ..   required_car_parking_spaces = col_double(),
##   ..   total_of_special_requests = col_double(),
##   ..   reservation_status = col_character(),
##   ..   reservation_status_date = col_date(format = "")
##   .. )
##  - attr(*, "problems")=<externalptr>
```

```
glimpse(bookings_df)
```

```
## Rows: 119,390
## Columns: 32
## $ hotel                          <chr> "Resort Hotel", "Resort Hotel", "Resort…
## $ is_canceled                    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, …
## $ lead_time                      <dbl> 342, 737, 7, 13, 14, 14, 0, 9, 85, 75, …
## $ arrival_date_year              <dbl> 2015, 2015, 2015, 2015, 2015, 2015, 201…
## $ arrival_date_month             <chr> "July", "July", "July", "July", "July",…
## $ arrival_date_week_number       <dbl> 27, 27, 27, 27, 27, 27, 27, 27, 27, 27,…
## $ arrival_date_day_of_month      <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, …
## $ stays_in_weekend_nights        <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, …
## $ stays_in_week_nights           <dbl> 0, 0, 1, 1, 2, 2, 2, 2, 3, 3, 4, 4, 4, …
## $ adults                         <dbl> 2, 2, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, …
## $ children                       <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, …
## $ babies                         <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, …
## $ meal                           <chr> "BB", "BB", "BB", "BB", "BB", "BB", "BB…
## $ country                        <chr> "PRT", "PRT", "GBR", "GBR", "GBR", "GBR…
## $ market_segment                 <chr> "Direct", "Direct", "Direct", "Corporat…
## $ distribution_channel           <chr> "Direct", "Direct", "Direct", "Corporat…
## $ is_repeated_guest              <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, …
## $ previous_cancellations         <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, …
## $ previous_bookings_not_canceled <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, …
## $ reserved_room_type             <chr> "C", "C", "A", "A", "A", "A", "C", "C",…
## $ assigned_room_type             <chr> "C", "C", "C", "A", "A", "A", "C", "C",…
## $ booking_changes                <dbl> 3, 4, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, …
## $ deposit_type                   <chr> "No Deposit", "No Deposit", "No Deposit…
## $ agent                          <chr> "NULL", "NULL", "NULL", "304", "240", "…
## $ company                        <chr> "NULL", "NULL", "NULL", "NULL", "NULL",…
## $ days_in_waiting_list           <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, …
## $ customer_type                  <chr> "Transient", "Transient", "Transient", …
## $ adr                            <dbl> 0.00, 0.00, 75.00, 75.00, 98.00, 98.00,…
## $ required_car_parking_spaces    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, …
## $ total_of_special_requests      <dbl> 0, 0, 0, 0, 1, 1, 0, 1, 1, 0, 0, 0, 3, …
## $ reservation_status             <chr> "Check-Out", "Check-Out", "Check-Out", …
## $ reservation_status_date        <date> 2015-07-01, 2015-07-01, 2015-07-02, 20…
```

We can also use `colnames()` to check the names of the columns in our data set.

```
colnames(bookings_df)
```

```
##  [1] "hotel"                          "is_canceled"
##  [3] "lead_time"                      "arrival_date_year"
##  [5] "arrival_date_month"             "arrival_date_week_number"
##  [7] "arrival_date_day_of_month"      "stays_in_weekend_nights"
##  [9] "stays_in_week_nights"           "adults"
## [11] "children"                       "babies"
## [13] "meal"                           "country"
## [15] "market_segment"                 "distribution_channel"
## [17] "is_repeated_guest"              "previous_cancellations"
## [19] "previous_bookings_not_canceled" "reserved_room_type"
## [21] "assigned_room_type"             "booking_changes"
## [23] "deposit_type"                   "agent"
## [25] "company"                        "days_in_waiting_list"
## [27] "customer_type"                  "adr"
## [29] "required_car_parking_spaces"    "total_of_special_requests"
## [31] "reservation_status"             "reservation_status_date"
```

Use the `skim_without_charts()` function from the `skimr` package:

```
skim_without_charts(bookings_df)
```

Data summary

Name	bookings_df
Number of rows	119390
Number of columns	32
_______________________	
Column type frequency:	
character	13
Date	1
numeric	18
________________________	
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
hotel	0	1	10	12	0	2	0
arrival_date_month	0	1	3	9	0	12	0
meal	0	1	2	9	0	5	0
country	0	1	2	4	0	178	0

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
market_segment	0	1	6	13	0	8	0
distribution_channel	0	1	3	9	0	5	0
reserved_room_type	0	1	1	1	0	10	0
assigned_room_type	0	1	1	1	0	12	0
deposit_type	0	1	10	10	0	3	0
agent	0	1	1	4	0	334	0
company	0	1	1	4	0	353	0
customer_type	0	1	5	15	0	4	0
reservation_status	0	1	7	9	0	3	0

Variable type: Date

skim_variable	n_missing	complete_rate	min	max	median	n_unique
reservation_status_date	0	1	2014-10-17	2017-09-14	2016-08-07	926

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
is_canceled	0	1	0.37	0.48	0.00	0.00	0.00	1	1
lead_time	0	1	104.01	106.86	0.00	18.00	69.00	160	737
arrival_date_year	0	1	2016.16	0.71	2015.00	2016.00	2016.00	2017	2017
arrival_date_week_number	0	1	27.17	13.61	1.00	16.00	28.00	38	53
arrival_date_day_of_month	0	1	15.80	8.78	1.00	8.00	16.00	23	31
stays_in_weekend_nights	0	1	0.93	1.00	0.00	0.00	1.00	2	19
stays_in_week_nights	0	1	2.50	1.91	0.00	1.00	2.00	3	50
adults	0	1	1.86	0.58	0.00	2.00	2.00	2	55
children	4	1	0.10	0.40	0.00	0.00	0.00	0	10
babies	0	1	0.01	0.10	0.00	0.00	0.00	0	10
is_repeated_guest	0	1	0.03	0.18	0.00	0.00	0.00	0	1
previous_cancellations	0	1	0.09	0.84	0.00	0.00	0.00	0	26
previous_bookings_not_canceled	0	1	0.14	1.50	0.00	0.00	0.00	0	72
booking_changes	0	1	0.22	0.65	0.00	0.00	0.00	0	21
days_in_waiting_list	0	1	2.32	17.59	0.00	0.00	0.00	0	391
adr	0	1	101.83	50.54	-6.38	69.29	94.58	126	5400
required_car_parking_spaces	0	1	0.06	0.25	0.00	0.00	0.00	0	8

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
total_of_special_requests	0	1	0.57	0.79	0.00	0.00	0.00	1	5

Step 4: Cleaning our data

Based on our notes we are primarily interested in the following variables: hotel, is_canceled, lead_time. Let's create a new data frame with just those columns, calling it `trimmed_df`.

```
trimmed_df <- bookings_df %>%
  select(hotel, is_canceled, lead_time)
```

Rename the variable 'hotel' to be named 'hotel_type' to be crystal clear on what the data is about:

```
trimmed_df %>%
  select(hotel, is_canceled, lead_time) %>%
  rename(hotel_type = hotel)
```

```
## # A tibble: 119,390 × 3
##    hotel_type   is_canceled lead_time
##    <chr>              <dbl>     <dbl>
##  1 Resort Hotel           0       342
##  2 Resort Hotel           0       737
##  3 Resort Hotel           0         7
##  4 Resort Hotel           0        13
##  5 Resort Hotel           0        14
##  6 Resort Hotel           0        14
##  7 Resort Hotel           0         0
##  8 Resort Hotel           0         9
##  9 Resort Hotel           1        85
## 10 Resort Hotel           1        75
## # ℹ 119,380 more rows
```

Now, we can combine the arrival month and year into one column using the unite() function:

```
comb_df <- bookings_df %>%
  select(arrival_date_year, arrival_date_month) %>%
  unite(arrival_month_year, c("arrival_date_month", "arrival_date_year"), sep = " ")
```

Step 5: Another way of doing things

We can also use the `mutate()` function to make changes to our columns. Let's say We wanted to create a new column that summed up all the adults, children, and babies on a reservation for the total number of people.

```
comb_df <- bookings_df %>%
  mutate(guests = adults + children + babies)

head(comb_df)
```

```
## # A tibble: 6 × 33
##   hotel        is_canceled lead_time arrival_date_year arrival_date_month
##   <chr>              <dbl>     <dbl>             <dbl> <chr>
## 1 Resort Hotel           0       342              2015 July
## 2 Resort Hotel           0       737              2015 July
## 3 Resort Hotel           0         7              2015 July
## 4 Resort Hotel           0        13              2015 July
## 5 Resort Hotel           0        14              2015 July
## 6 Resort Hotel           0        14              2015 July
## # ℹ 28 more variables: arrival_date_week_number <dbl>,
## #   arrival_date_day_of_month <dbl>, stays_in_weekend_nights <dbl>,
## #   stays_in_week_nights <dbl>, adults <dbl>, children <dbl>, babies <dbl>,
## #   meal <chr>, country <chr>, market_segment <chr>,
## #   distribution_channel <chr>, is_repeated_guest <dbl>,
## #   previous_cancellations <dbl>, previous_bookings_not_canceled <dbl>,
## #   reserved_room_type <chr>, assigned_room_type <chr>, …
```

Great. Now it's time to calculate some summary statistics! Calculate the total number of canceled bookings and the average lead time for booking - We'll want to start our code after the %>% symbol. Make a column called 'number_canceled' to represent the total number of canceled bookings. Then, make a column called 'average_lead_time' to represent the average lead time.

```
comb_df <- bookings_df %>%
  summarize(number_canceled = sum(is_canceled),
            average_lead_time = mean(lead_time))

head(comb_df)
```

```
## # A tibble: 1 × 2
##   number_canceled average_lead_time
##             <dbl>             <dbl>
## 1           44224              104.
```

Activity Wrap Up

We cleaned and analyzd data in `R`; We used basic cleaning functions like `rename()` and `clean_names()` and performed basic calculations on real data.